

Lab 10 Practice

October 2025

1 BCD Synchronous Counter

- a. [Theory] Design a 4-bit **BCD Synchronous Counter** using JK flip-flops to count from 0000 (0) to 1001 (9) and then reset to 0000. Draw the state table, derive JK excitation equations, and sketch the logic diagram showing synchronous operation.
- b. [Verilog] Write a Verilog code for the 4-bit **BCD Synchronous Counter** using JK flip-flops. Include a synchronous reset and verify the count sequence:

$$0000 \rightarrow 0001 \rightarrow \dots \rightarrow 1001 \rightarrow 0000$$

2 Custom Sequence Counter using T Flip-Flops

- a. [Theory] Design a **Synchronous Sequence Counter** using T flip-flops that follows the sequence

$$0 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 0 \rightarrow 2 \rightarrow \dots$$

Represent each state in binary and construct the **state table** showing present and next states. Derive the required **T flip-flop excitation equations** using Karnaugh maps and draw the logic diagram showing the connection of T inputs with logic gates.

- b. [Verilog] Write a Verilog module to implement the above sequence counter using **T flip-flops**. The design should include:
- A **tff** module describing the T flip-flop behavior.
 - A top-level **seq_counter** module that generates the sequence.
 - A synchronous reset input to initialize the counter to state 000.

Simulate the design to verify the output sequence:

$$000 \rightarrow 010 \rightarrow 011 \rightarrow 101 \rightarrow 110 \rightarrow 000 \rightarrow 010 \rightarrow \dots$$